

Bayesian Computational Statistics

Bayes' Theorem, Rejection Sampling,
Importance Sampling and Importance Resampling

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1 Bayes' Theorem and Bayesian Inference

Let θ denote the parameters of interest and y denote the observed data.

Bayes' Theorem [1, Ch. 1] combines the prior distribution $p(\theta)$ with the sampling model (likelihood) $p(y | \theta)$ to obtain the posterior distribution $p(\theta | y)$:

$$p(\theta | y) = \frac{p(y | \theta) p(\theta)}{p(y)} = \frac{p(y | \theta) p(\theta)}{\int p(y | \theta) p(\theta) d\theta} \quad (1)$$

Where:

- $p(\theta)$ is the **prior distribution**, representing belief before seeing the data.
- $p(y | \theta)$ is the **likelihood function** when viewed as a function of θ .
- $p(\theta | y)$ is the **posterior distribution**.
- $p(y) = \int p(y | \theta) p(\theta) d\theta$ is the **marginal likelihood** (or evidence), which acts as a normalizing constant.

In practice, $p(y)$ is often intractable, so we work with the unnormalized posterior density:

$$p(\theta | y) \propto q(\theta | y) = p(y | \theta) p(\theta) \quad (2)$$

2 Direct Simulation and Rejection Sampling

When independent samples cannot be drawn directly from $p(\theta | y)$, we use proposal distributions [1, Sec. 10.3].

2.1 Rejection Sampling

To draw samples from an unnormalized target density $f(\theta) \propto p(\theta | y)$, choose a proposal distribution $g(\theta)$ such that for some known constant $M > 0$:

$$f(\theta) \leq M g(\theta) \quad \text{for all } \theta \text{ where } f(\theta) > 0. \quad (3)$$

Algorithm 1 Rejection Sampling

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1: repeat
2:   Sample a candidate  $\theta^* \sim g(\theta)$ .
3:   Sample  $U \sim \text{Uniform}(0, 1)$ .
4:   Calculate acceptance ratio  $r = \frac{f(\theta^*)}{M g(\theta^*)}$ .
5: until  $U \leq r$ 
6: return  $\theta = \theta^*$  as a draw from  $p(\theta | y)$ .
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Properties and limitations.

- The accepted points θ are independent draws from the exact target distribution.
- Expected acceptance rate is $1/M$.
- Finding an efficient M becomes exponentially difficult in high dimensions.

3 Importance Ratio and Importance Sampling

Importance sampling [1, Sec. 10.4] is a variance-reduction Monte Carlo method used to compute expectations of functions $h(\theta)$ with respect to the target distribution $p(\theta | y)$ without requiring exact draws.

3.1 Importance Ratio

If target distribution $p(\theta | y) \propto q(\theta | y)$ and proposal distribution $g(\theta)$ are used, the **importance ratio** (or raw importance weight) for a point θ^s is:

$$w(\theta^s) = \frac{p(\theta^s | y)}{g(\theta^s)} \quad \text{or, using unnormalized densities,} \quad w(\theta^s) = \frac{q(\theta^s | y)}{g(\theta^s)} \quad (4)$$

3.2 Importance Sampling Expectation Estimator

To estimate $\mathbb{E}[h(\theta) | y] = \int h(\theta)p(\theta | y)d\theta$:

Draw S independent samples $\theta^1, \theta^2, \dots, \theta^S \sim g(\theta)$ and calculate normalized weights:

$$\tilde{w}^s = \frac{w(\theta^s)}{\sum_{j=1}^S w(\theta^j)} \quad (5)$$

The importance sampling estimator is given by:

$$\hat{\mathbb{E}}[h(\theta) | y] = \sum_{s=1}^S \tilde{w}^s h(\theta^s) = \frac{\sum_{s=1}^S w(\theta^s) h(\theta^s)}{\sum_{s=1}^S w(\theta^s)} \quad (6)$$

Key considerations.

- Requires $g(\theta) > 0$ whenever $p(\theta | y) > 0$.
- The tail behavior of $g(\theta)$ should be heavier than $p(\theta | y)$; otherwise, $w(\theta)$ can have unbounded variance.

4 Sampling-Importance Resampling / SIR

Sampling-Importance Resampling (also known as Importance Resampling) [1, Sec. 10.4] is used to draw *unweighted* approximate samples from the target distribution $p(\theta | y)$.

4.1 SIR Algorithm

To obtain k draws from $p(\theta | y)$:

Algorithm 2 Sampling-Importance Resampling (SIR)

- 1: Generate S candidate draws $\theta^1, \theta^2, \dots, \theta^S$ independently from proposal distribution $g(\theta)$ (where $S \gg k$).
- 2: Calculate raw weights $w(\theta^s) = \frac{q(\theta^s | y)}{g(\theta^s)}$ for $s = 1, \dots, S$.
- 3: Compute normalized probabilities:

$$p_s = \tilde{w}^s = \frac{w(\theta^s)}{\sum_{j=1}^S w(\theta^j)}$$

- 4: Resample k draws from candidate pool $\{\theta^1, \dots, \theta^S\}$ with replacement according to probabilities $(p_1, \dots, p_S)$.
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4.2 Comparison

Method	Output Type	Proposal Requirements	Main Drawback
Rejection Sampling	Exact independent draws	Requires bounding $M \cdot g(\theta) \geq f(\theta)$	Inefficient in high dimensions
Importance Sampling	Weighted samples	Requires $g(\theta)$ with heavier tails	Estimates expectations only
Importance Resampling	Approximate unweighted draws	$S \gg k$, flexible proposal $g(\theta)$	Dependent draws if k/S too large

References

- [1] Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2013). *Bayesian Data Analysis*, 3rd edition. Chapman & Hall/CRC, Boca Raton, FL.